Multi-Factor Models & Performance Eval

- high book-to-market (B/M) firms (value firms) have historically performed better than low B/M firms (growth firms).
  - So over the course of 100 years, you would've 10x'ed your return by investing in asset holding firms vs growth (like Apple)
- value firms do better on average, but when they do poorly, they all do poorly at the same time

- Fama-French 3 factor model (FF3): $E(r_i) - r_f = B_{m,i} E(r_m - r_f) + B_{SMB} E(r_m - r_f) + B_{HML} E[r_m - r_f]$
- portfolio invested 40% in A and 60% in B

Factor	Beta A	Beta B
Market	0.8	1.4
SMB	-0.4	0.6
HML	1.2	0

exposure to market $= 0.4(0.8) + 0.6(1.4) = 1.16$

exposure to SMB $= \dots = 0.2$

exposure to HML $= \dots = 0.48$

Given: E[r] of market is 7%, SMB is 4%, HML is 6%

E(excess return of portfolio) $= 1.16(0.07) + 0.2(0.04) + 0.48(0.06) = 0.118$